

General Knowledge

Indian Geography

STUDY NOTES

Wildlife Sanctuaries and Biosphere Reserves



WILDLIFE SANCTUARIES AND BIOSPHERE RESERVES

India is renowned for its vast **forests** and **diverse wildlife**. A **wildlife sanctuary** is a **protected area** established to conserve **plants** and **animals**. **Tourism** is not allowed in these sanctuaries, as they serve as **natural habitats**. The **Wildlife Protection Act of 1947** created these protected areas. Wildlife sanctuaries in India are classified under **IUCN Category IV** as **protected areas**. The **Wildlife (Protection) Act of 1972** empowered state governments to declare specific regions as wildlife sanctuaries if they were deemed of **ecological**, **geomorphologic**, and **natural value**.

As of November 2023, India has a total of **573 wildlife sanctuaries**, covering **123,762.56 km²**, or **3.76%** of the country's land area. Additionally, **218 more sanctuaries** are proposed in the **Protected Area Network Report**, covering an area of **16,829 km²**.



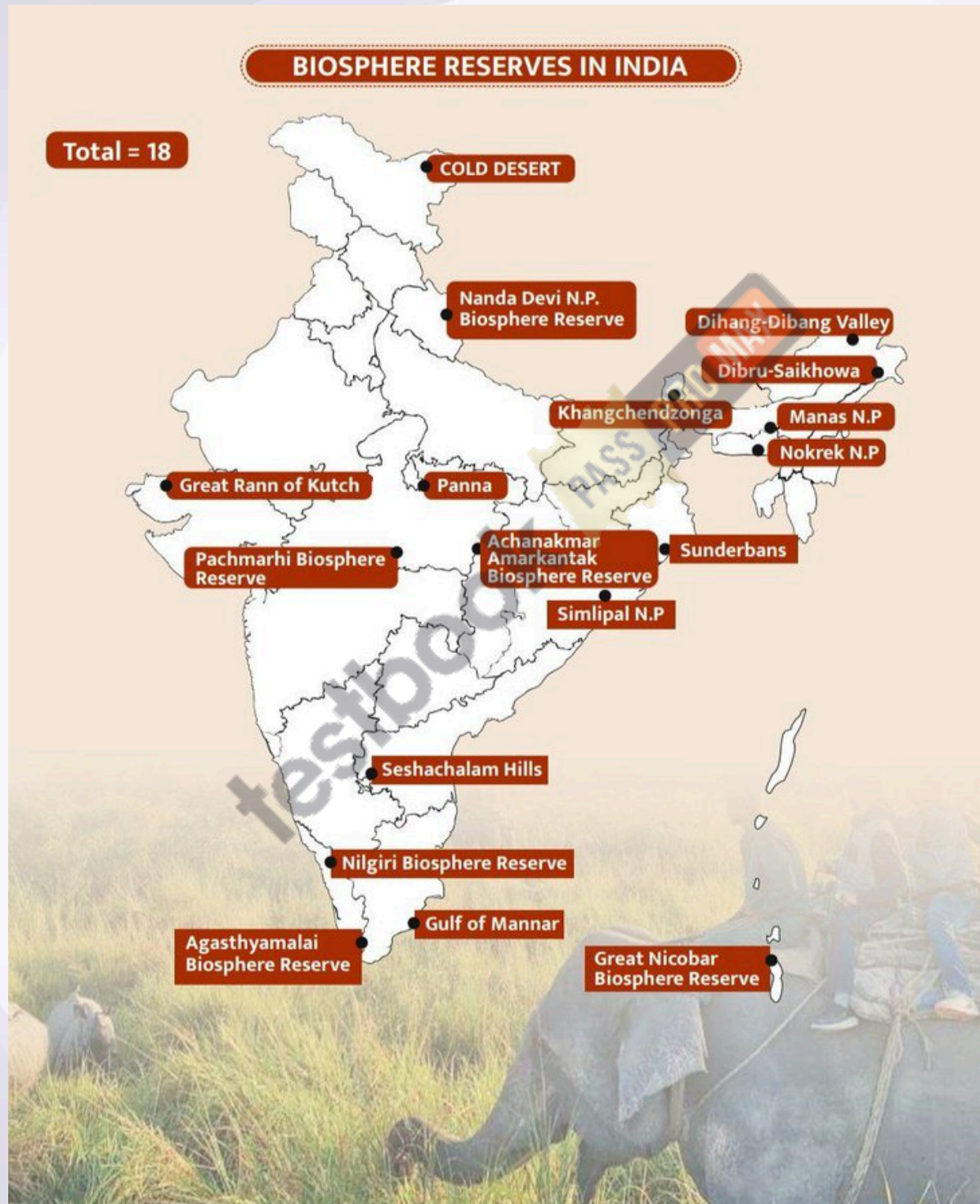
WILDLIFE SANCTUARIES OF INDIA STATES/UT

States/Union Territories	Wildlife Sanctuaries
Assam	Nambor WLS, Dihing Patkai WLS, East Karbi Anglong WLS, Chakrashila WLS, Amchang WLS
Bihar	Kaimur WLS, Gautam Budha WLS, Pant (Rajgir) WLS, Valmiki WLS
Chhattisgarh	Bhairamgarh WLS, Badalkhol WLS, Bhoramdev WLS, Udanti Wild Buffalo WLS
Goa	Bondla WLS, Madei WLS
Gujarat	Kutch Desert WLS, Porbandar Lake WLS, Jambugodha WLS, Wild Ass WLS, Ratanmahal WLS, Thol Lake WLS, Sasan Gir Sanctuary, Mitiyala WLS
Haryana	Bhindawas WLS, N Khaparwas WLS, Kalesar WLS
Himachal Pradesh	Bandli WLS, Daranghati WLS, Dhauladhar WLS, Talra WLS, Pong Dam Lake WLS, Nargu WLS
Jharkhand	Lawalong WLS, Parasnath WLS, Palkot WLS
Karnataka	Someshwara WLS, Bhadra WLS, Bhimgad WLS, Brahmagiri WLS, Cauvery WLS, Pushpagiri WLS, Sharavathi Valley WLS
Kerala	Periyar Wildlife Sanctuary, Chinnar Wildlife Sanctuary, Aralam WLS, Chimmony WLS, Idukki WLS, Malabar WLS
Madhya Pradesh	Bori WLS, Gandhi Sagar WLS, Ken Gharial WLS, National Chambal WLS, Orcha WLS
Maharashtra	Koyana WLS, Painganga WLS, Bhimashankar WLS, Tungareshwar WLS, Great Indian Bustard WLS
Manipur	Yangoupokpi-Lokchao WLS
Meghalaya	—
Mizoram	Dampa WLS (TR), Ngengpui WLS, Baghmara Pitcher Plant WLS
Nagaland	Fakim WLS, Rangapahar WLS
Odisha	Baisipalli WLS, Chilika (Nalaban) WLS, Hadgarh WLS, Satkosia Gorge WLS
Punjab	Abohar WLS, Harike Lake WLS, Jhajjar Bacholi WLS
Rajasthan	Keoladeo Bird Sanctuary, Jawahar Sagar WLS, Mount Abu WLS, Ramsagar WLS, Shergarh WLS
Sikkim	Fambong Lho WLS, Kitam WLS (Bird), Maenam WLS

Tamil Nadu	Indira Gandhi (Annamalai) WLS, Karaivetti WLS, Pulicat Lake WLS, Vedanthangal WLS, Kalakad WLS
Tripura	Gumti WLS, Rowa WLS, Trishna WLS
Uttarakhand	Askot Musk Deer WLS, Binsar WLS, Govind Pashu Vihar WLS, Kedarnath WLS, Sonanadi WLS
Uttar Pradesh	Hastinapur WLS, Ranipur WLS, Sohagibarwa WLS, Sur Sarovar WLS, Chandraprabha WLS, National Chambal WLS
West Bengal	Sunderbans Wildlife Sanctuary, Chintamani Kar Bird Sanctuary, Haliday Island WLS, Ballavpur WLS, Lothian Island WLS, Mahananda WLS
Andaman and Nicobar Islands	Bamboo Island WLS, Barren Island WLS, Chanel Island WLS, Peacock Island WLS, Turtle Islands WLS
Jammu & Kashmir	Gulmarg WLS, Limber WLS, Nandini WLS
Lakshadweep	Pitti WLS (Bird)
Dadra Nagar Haveli and Daman and Diu	Dadra & Nagar Haveli WLS, Fudam WLS

- **Largest Wildlife Sanctuary:** Kutch Desert Wildlife Sanctuary, Gujarat (7,505 km²), home to the Indian Wild Ass.
- **Smallest Wildlife Sanctuary:** Sitamata Wildlife Sanctuary, Rajasthan (423.1 km²), known for Leopards and Chinkaras.
- **Oldest Wildlife Sanctuary:** Vedanthangal Bird Sanctuary, Tamil Nadu (established 1796), hosting thousands of migratory birds.
- **Newest Wildlife Sanctuary:** Madhav National Park, Madhya Pradesh (established 1959), known for Tigers and Leopards.
- **Largest Tiger Reserve:** Sundarbans Tiger Reserve, West Bengal (2,585 km²), famous for the Royal Bengal Tiger.
- **Smallest Tiger Reserve:** Madhav National Park, Madhya Pradesh (375 km²), home to Tigers and Leopards.
- **Largest Bird Sanctuary:** Keoladeo National Park (Bharatpur Bird Sanctuary), Rajasthan (29 km²), hosting over 370 species of birds.
- **Largest Marine Sanctuary:** Rani Jhansi Marine National Park, Andaman and Nicobar Islands (256 km²), rich in coral reefs and marine life.
- **Highest Wildlife Sanctuary:** Khangchendzonga National Park, Sikkim (1,829m to 8,586m), home to Snow Leopards and Red Pandas.
- **Most Ecologically Diverse Sanctuary:** Periyar Wildlife Sanctuary, Kerala (925 km²), known for Elephants, Tigers, and Nilgiri Langurs.

Biosphere reserves are large areas that include **national parks** and **animal wildlife sanctuaries**. These build a high ratio of **natural habitat** with constitutional protection under the particular **protection policies** and arrangements. As of 2024, **India** has **18 biosphere reserves**, **12** of which are part of the **UNESCO's Man and Biosphere Reserve (MAB) Programme**.

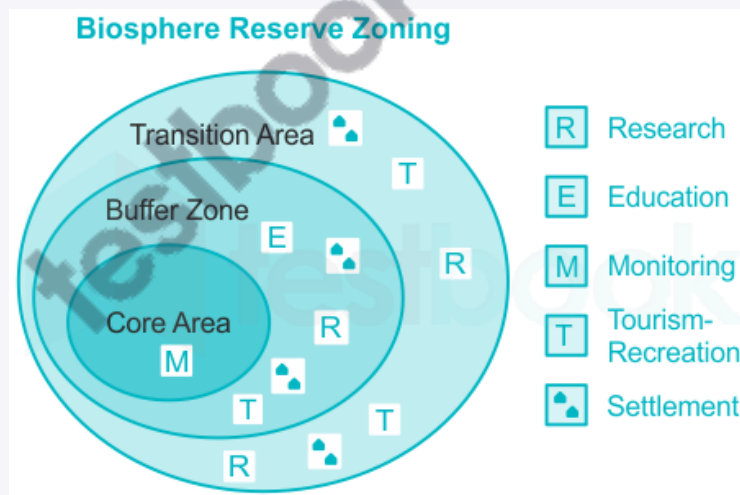


Criteria for Designation of Biosphere Reserves:

1. **Effective Protection of Core Area:** The core area must be rigorously protected, with minimal human disturbance.
2. **Ecological Integrity:** The core area should be a distinct biogeographical unit, large enough to sustain a viable population of species across all trophic levels within the ecosystem.
3. **Biodiversity Conservation:** The site must be of significant value for the conservation of biological diversity.
4. **Research and Sustainable Practices:** The area should include additional land and water resources suitable for conducting research and demonstrating sustainable management practices.
5. **Cultural and Traditional Preservation:** The area should support the preservation of traditional tribal or rural lifestyles, fostering a harmonious relationship with the environment.

Zonification of Biosphere Reserves in India

Biosphere reserves are divided into three interrelated zones, each serving distinct functions:



1. Core Zone

The **core zone** is a **strictly protected area** within the biosphere reserve, governed by the **Wildlife (Protection) Act 1972**. Its primary role is the **conservation of landscapes, ecosystems, species, and genetic diversity**. This zone is off-limits to human activities that could disrupt the natural processes or harm wildlife.

2. Buffer Zone

The **buffer zone** surrounds or adjoins the core zone. This area allows for activities like **scientific research, monitoring, education, and training**. Human activities that do not negatively impact the **natural ecosystem** of the biosphere reserve are permitted here, serving as a transition between the core and more human-influenced areas.

3. Transition Zone

The **transition zone** is the outermost region of the biosphere reserve. It supports **sustainable economic and human activities**, including **agriculture and settlements**. This zone enables local communities to engage in practices that are compatible with conservation efforts.

BIOSPHERE RESERVE OF INDIA

Name	Year of Notification	Location
Nilgiri Biosphere Reserve	1986	States: Tamil Nadu, Kerala, Karnataka Locations: Wayanad, Nagarhole, Bandipur, Madumalai, Nilambur, Silent Valley, Siruvani Hills
Nanda Devi Biosphere Reserve	1988	State: Uttarakhand Locations: Chamoli, Pithoragarh, Almora districts
Nokrek Biosphere Reserve	1988	State: Meghalaya Locations: East, West, and South Garo Hills districts
Great Nicobar Biosphere Reserve	1989	State: Andaman and Nicobar Islands Location: Southernmost region of the island
Gulf of Mannar Biosphere Reserve	1989	State: Tamil Nadu Locations: From Rameswaram island (North) to Kanyakumari (South)
Manas Biosphere Reserve	1989	State: Assam Locations: Kokrajhar, Bongaigaon, Barpeta, Nalbari, Kamrup, Darang districts
Sunderbans Biosphere Reserve	1989	State: West Bengal Location: Delta of Ganges and Brahmaputra river system
Simlipal Biosphere Reserve	1994	State: Odisha Location: Mayurbhanj district
Dibru – Saikhowa Biosphere Reserve	1997	State: Assam Locations: Dibrugarh and Tinsukia districts
Dehang-Dibang Biosphere Reserve	1998	State: Arunachal Pradesh Locations: Upper Siang, West Siang, Dibang Valley districts
Pachmarhi Biosphere Reserve	1999	State: Madhya Pradesh Locations: Betul, Hoshangabad, Chhindwara districts

Khangchendzonga Biosphere Reserve	2000	State: Sikkim Locations: North and West districts
Agasthyamalai Biosphere Reserve	2001	States: Tamil Nadu, Kerala Locations: Tirunelveli, Kanyakumari (TN); Thiruvananthapuram, Kollam, Pathanamthitta (Kerala)
Achanakmar-Amarkantak Biosphere Reserve	2005	States: Madhya Pradesh, Chhattisgarh Locations: Anuppur, Dindori (MP); Bilaspur (CG)
Kachchh Biosphere Reserve	2008	State: Gujarat Locations: Kachchh, Rajkot, Surendranagar, Patan districts
Cold Desert Biosphere Reserve	2009	State: Himachal Pradesh Locations: Pin Valley NP, Chandratat, Sarchu, Kibber wildlife sanctuary
Seshachalam Hills Biosphere Reserve	2010	State: Andhra Pradesh Locations: Chittoor and Kadapa districts in the Eastern Ghats
Panna Biosphere Reserve	2011	State: Madhya Pradesh Locations: Panna, Chhatarpur districts

UNESCO BIOSPHERE RESERVE OF INDIA

Sl. No.	Name	Year	State
1	Nilgiri Biosphere Reserve	2000	Tamil Nadu, Kerala, and Karnataka
2	Gulf of Mannar Biosphere Reserve	2001	Tamil Nadu
3	Sunderbans Biosphere Reserve	2001	West Bengal
4	Nanda Devi Biosphere Reserve	2004	Uttarakhand
5	Similipal Biosphere Reserve	2009	Odisha
6	Pachmarhi Biosphere Reserve	2009	Madhya Pradesh
7	Nokrek Biosphere Reserve	2009	Meghalaya
8	Achanakmar-Amarkantak Biosphere Reserve	2012	Madhya Pradesh and Chhattisgarh

9	Great Nicobar Biosphere Reserve	2013	Andaman and Nicobar Islands
10	Agasthyamalai Biosphere Reserve	2016	Tamil Nadu and Kerala
11	Khangchendzonga Biosphere Reserve	2018	Sikkim
12	Panna Biosphere Reserve	2020	Madhya Pradesh

- **Largest Biosphere Reserve: Gulf of Mannar** (10,500 km²) in Tamil Nadu, known for its rich marine biodiversity.
- **Smallest Biosphere Reserve: Nokrek** (820 km²) in Meghalaya, home to the red panda and tropical rainforests.
- **Highest Biosphere Reserve: Khangchendzonga** in Sikkim, spanning from 1,829 m to 8,586 m, with diverse alpine and subtropical ecosystems.
- **Lowest Biosphere Reserve: Great Nicobar** in Andaman and Nicobar Islands, located at sea level, rich in unique mangrove and tropical ecosystems.
- **Most Biodiverse Biosphere Reserve: Nilgiri** (5,520 km²) in Tamil Nadu, Kerala, and Karnataka, housing elephants, tigers, and rare flora.
- **Most Recent Biosphere Reserve: Panna** (2020) in Madhya Pradesh, known for its thriving tiger population and waterfalls.
- **Largest Marine Biosphere Reserve: Gulf of Mannar** (10,500 km²), India's largest marine ecosystem, with coral reefs and sea turtles.
- **Richest in Cultural Heritage: Sunderbans** in West Bengal, a UNESCO World Heritage Site, is home to the Royal Bengal Tiger and unique wetlands.
- **Unique Wetland Ecosystem: Sunderbans** in West Bengal, is renowned for its mangrove forests and aquatic biodiversity.

Difference between Wildlife Sanctuaries and Biosphere Reserves

Aspect	Wildlife Sanctuary	Biosphere Reserve
Scope of Protection	Protects primarily wild animals	Protects entire ecosystems, including plants and animals
Human Activity	Allows some activities with minor restrictions	Stricter regulations; core areas have no human activities
Boundary Definition	Boundaries are often not fixed	Boundaries are fixed and designated by the government
Ownership	It can be government or privately owned	Generally owned and managed by government authorities
Size and Area	Typically covers a smaller area	Generally encompasses a larger area to maintain ecological diversity

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